



NSW Education Standards Authority

2018 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks** (pages 2–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8–19)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which expression is equal to $\int \frac{1}{\sqrt{1-4x^2}} dx$?

A. $\frac{1}{2} \sin^{-1} \frac{x}{2} + C$

B. $\frac{1}{2} \sin^{-1} 2x + C$

C. $\sin^{-1} \frac{x}{2} + C$

D. $\sin^{-1} 2x + C$

2 What are the equations of the asymptotes of the hyperbola $9x^2 - 4y^2 = 36$?

A. $y = \pm \frac{9}{4}x$

B. $y = \pm \frac{2}{3}x$

C. $y = \pm \frac{3}{2}x$

D. $y = \pm \frac{4}{9}x$

3 The cubic equation $x^3 + 2x^2 + 5x - 1 = 0$ has roots α , β and γ .

Which cubic equation has roots $\frac{-1}{\alpha}$, $\frac{-1}{\beta}$, $\frac{-1}{\gamma}$?

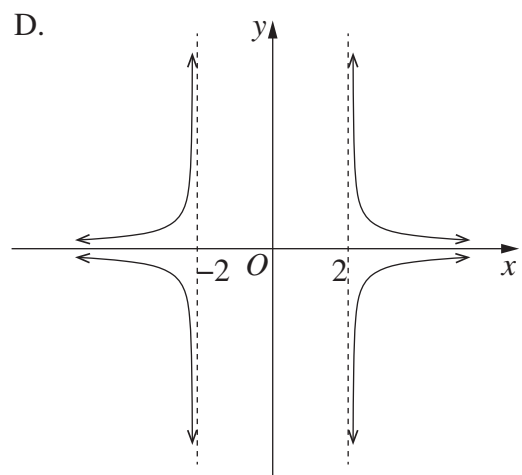
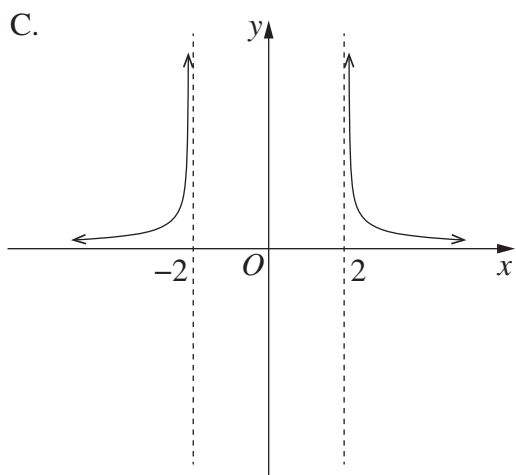
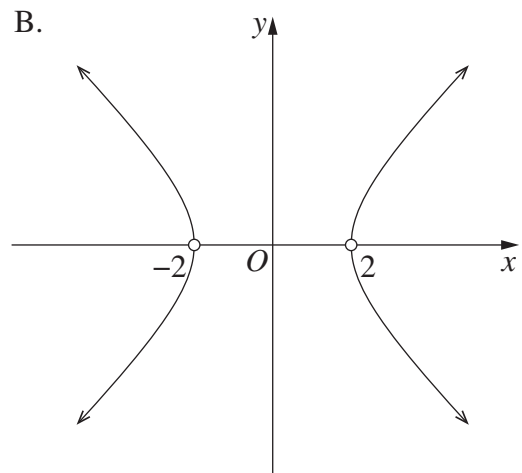
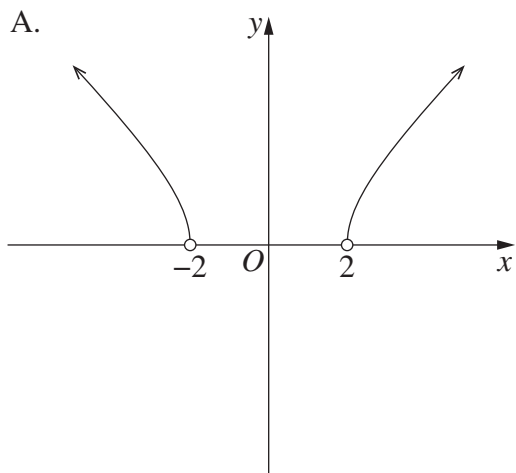
A. $x^3 - 5x^2 - 2x + 1 = 0$

B. $x^3 - 5x^2 - 2x - 1 = 0$

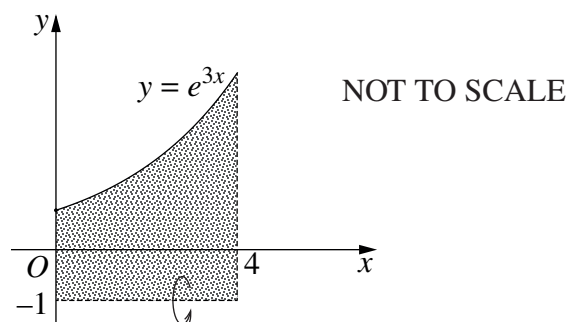
C. $x^3 + 5x^2 + 2x + 1 = 0$

D. $x^3 + 5x^2 - 2x + 1 = 0$

4 Which graph best represents the curve $y = \frac{1}{\sqrt{x^2 - 4}}$?



- 5 The diagram shows the graph $y = e^{3x}$ for $0 \leq x \leq 4$. The region bounded by $y = -1$, $y = e^{3x}$, $x = 0$ and $x = 4$ is rotated about the line $y = -1$ to form a solid.



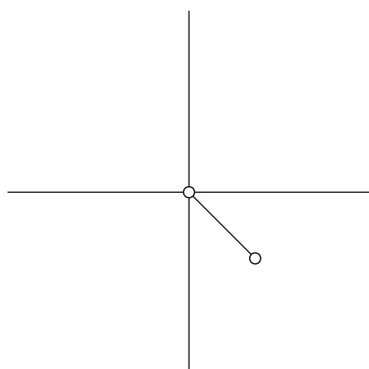
Which integral represents the volume of the solid formed?

- A. $\pi \int_0^4 (e^{3x} + 1)^2 dx$
- B. $2\pi \int_0^4 x(e^{3x} + 1) dx$
- C. $\pi \int_0^4 (e^{3x} - 1)^2 dx$
- D. $2\pi \int_0^4 x(e^{3x} - 1) dx$
- 6 Which complex number is a 6th root of i ?

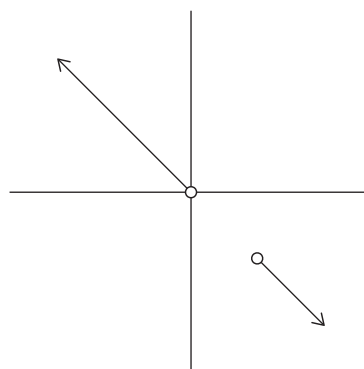
- A. $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$
- B. $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$
- C. $-\sqrt{2} + \sqrt{2}i$
- D. $-\sqrt{2} - \sqrt{2}i$

7 Which diagram best represents the solutions to the equation $\arg(z) = \arg(z + 1 - i)$?

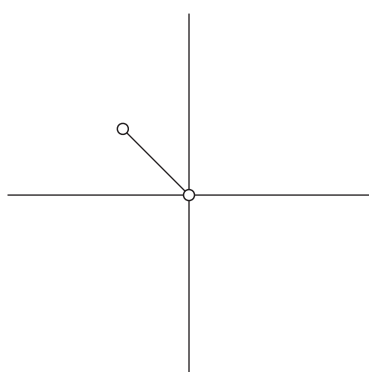
A.



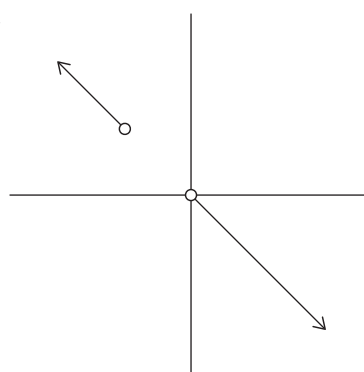
B.



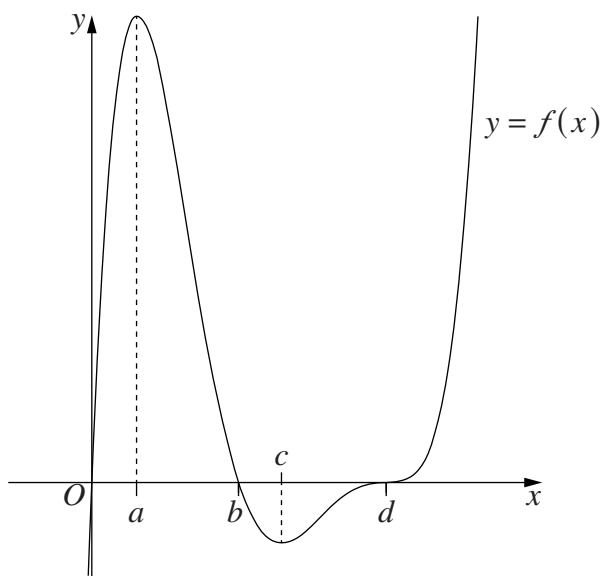
C.



D.



- 8 The diagram shows the graph of the curve $y = f(x)$.



Let $F(x) = \int_0^x f(t) dt$.

At what value(s) of x does the concavity of the curve $y = F(x)$ change?

- A. d
 - B. a, c
 - C. b, d
 - D. a, c, d
- 9 It is given that a, b are real and p, q are purely imaginary.

Which pair of inequalities must always be true?

- A. $a^2 p^2 + b^2 q^2 \leq 2abpq$, $a^2 b^2 + p^2 q^2 \leq 2abpq$
- B. $a^2 p^2 + b^2 q^2 \leq 2abpq$, $a^2 b^2 + p^2 q^2 \geq 2abpq$
- C. $a^2 p^2 + b^2 q^2 \geq 2abpq$, $a^2 b^2 + p^2 q^2 \leq 2abpq$
- D. $a^2 p^2 + b^2 q^2 \geq 2abpq$, $a^2 b^2 + p^2 q^2 \geq 2abpq$

- 10** Consider the functions $f(x) = \sin x$ and $g(x) = x \sin x$.

The x -coordinate of each stationary point of $f(x)$ is very close to the x -coordinate of a stationary point of $g(x)$.

Suppose $f(x)$ has a stationary point at $x = a$ and the stationary point of $g(x)$ with x -coordinate closest to $x = a$ is at $x = b$.

Which statement is always true?

- A. $a < b$
- B. $a > b$
- C. $|a| < |b|$
- D. $|a| > |b|$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Let $z = 2 + 3i$ and $w = 1 - i$.

(i) Find zw . **1**

(ii) Express $\bar{z} - \frac{2}{w}$ in the form $x + iy$, where x and y are real numbers. **2**

(b) The polynomial $p(x) = x^3 + ax^2 + b$ has a zero at r and a double zero at 4. **3**

Find the values of a , b and r .

(c) By writing $\frac{x^2 - x - 6}{(x+1)(x^2 - 3)}$ in the form $\frac{a}{x+1} + \frac{bx+c}{x^2 - 3}$, **4**

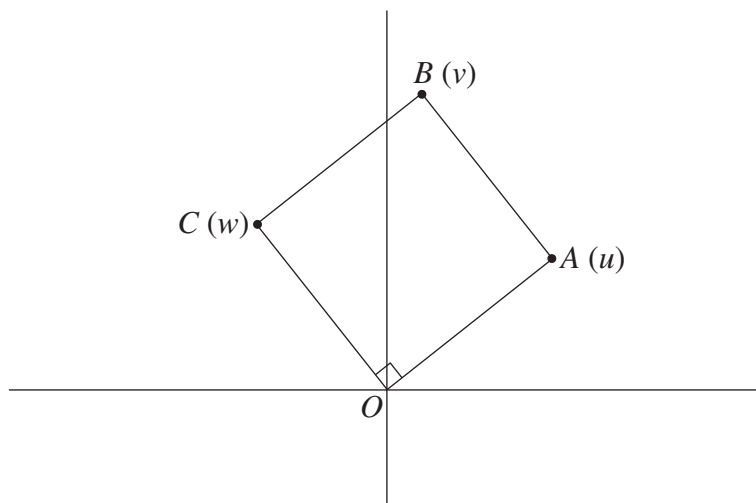
find $\int \frac{x^2 - x - 6}{(x+1)(x^2 - 3)} dx$.

Question 11 continues on page 9

Question 11 (continued)

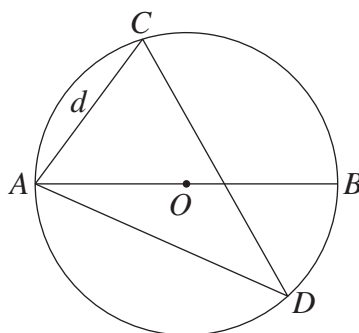
- (d) The points A , B and C on the Argand diagram represent the complex numbers u , v and w respectively.

The points O , A , B and C form a square as shown on the diagram.



It is given that $u = 5 + 2i$.

- | | |
|---|---|
| (i) Find w . | 1 |
| (ii) Find v . | 1 |
| (iii) Find $\arg\left(\frac{w}{v}\right)$. | 1 |
- (e) The circle centred at O with radius r has a diameter AB . Points C and D are chosen on the circle as shown in the diagram. The chord AC has length d . 2

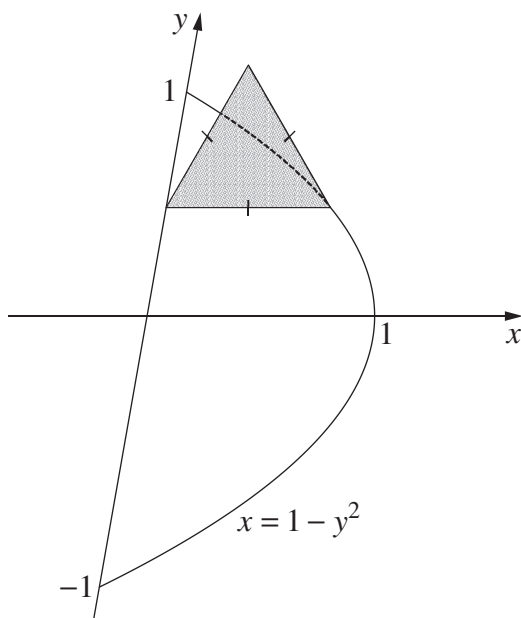


Show that $d = 2r \sin D$.

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) The base of a solid is the region enclosed by the parabola $x = 1 - y^2$ and the y -axis. Each cross-section perpendicular to the y -axis is an equilateral triangle, as shown in the diagram. **3**



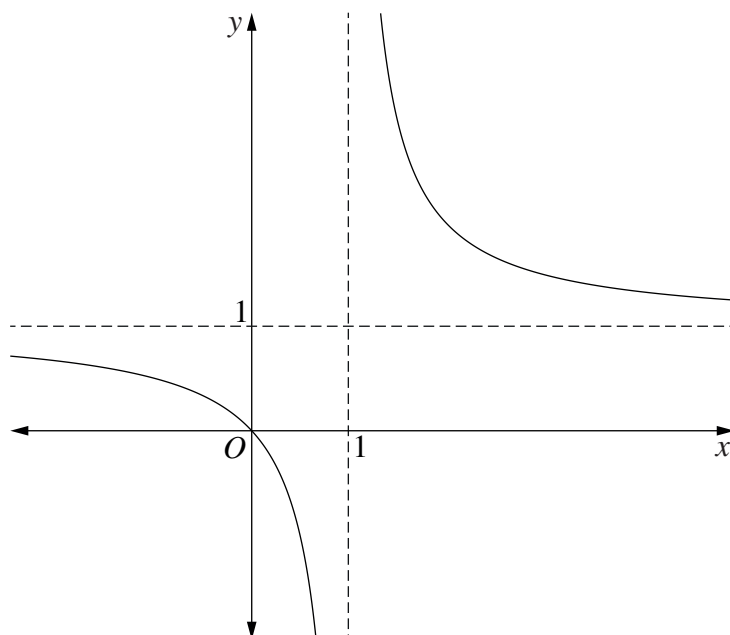
Find the volume of the solid.

- (b) A curve has equation $x^2 + xy + y^2 = 3$.
- (i) Use implicit differentiation to show that $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$. **2**
- (ii) Hence, or otherwise, find the coordinates of the points on the curve where $\frac{dy}{dx} = 0$. **2**
- (c) Find $\int \frac{x^2 + 2x}{x^2 + 2x + 5} dx$. **3**

Question 12 continues on page 11

Question 12 (continued)

- (d) The diagram shows the graph of the function $f(x) = \frac{x}{x-1}$.



Draw a separate half-page graph for each of the following functions, showing all asymptotes and intercepts.

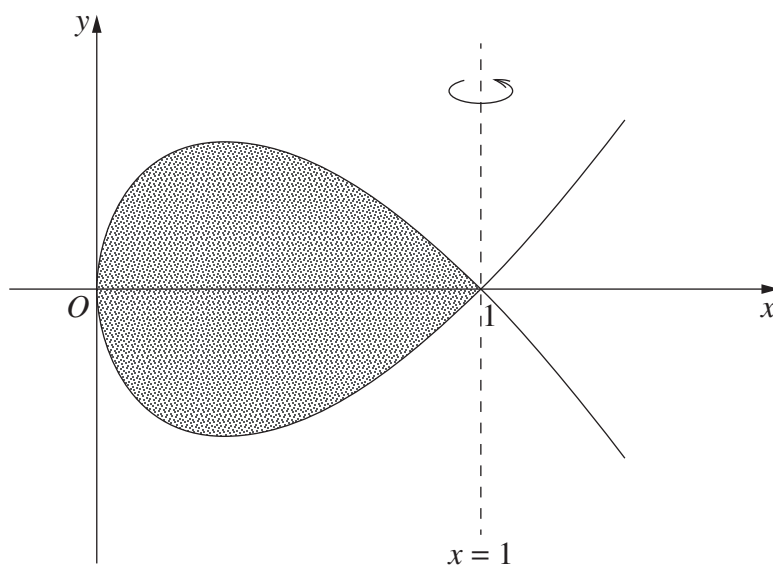
- | | |
|----------------------|----------|
| (i) $y = f(x) $ | 1 |
| (ii) $y = (f(x))^2$ | 2 |
| (iii) $y = x + f(x)$ | 2 |

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) The graph $y^2 = x(1-x)^2$ is shown.

3



Use the method of cylindrical shells to find the volume of the solid formed when the shaded region is rotated about the line $x = 1$.

- (b) Let $z = 1 - \cos 2\theta + i \sin 2\theta$, where $0 < \theta \leq \pi$.

(i) Show that $|z| = 2 \sin \theta$.

2

(ii) Show that $\arg(z) = \frac{\pi}{2} - \theta$.

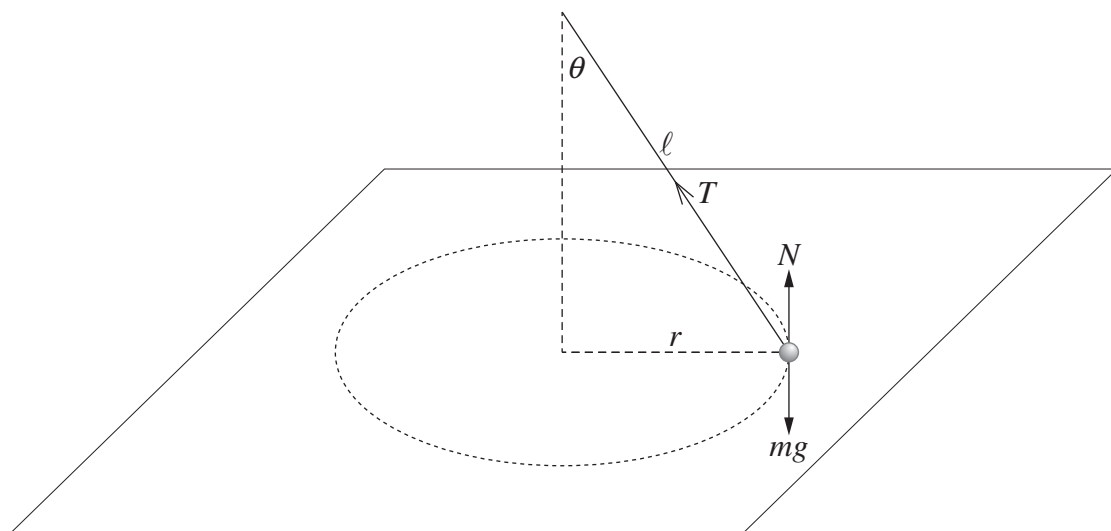
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Question 13 continues on page 13

Question 13 (continued)

- (c) A particle of mass m is attached to a light inextensible string of length ℓ . The string makes an angle θ to the vertical. **3**

The particle is moving in a circle of radius r on a smooth horizontal surface. The particle is moving with uniform angular velocity ω . The forces on the particle are the tension T in the string; the normal reaction N to the horizontal surface; and the gravitational force mg .



The particle remains in contact with the horizontal surface.

By resolving the forces on the particle in the horizontal and vertical directions, show that

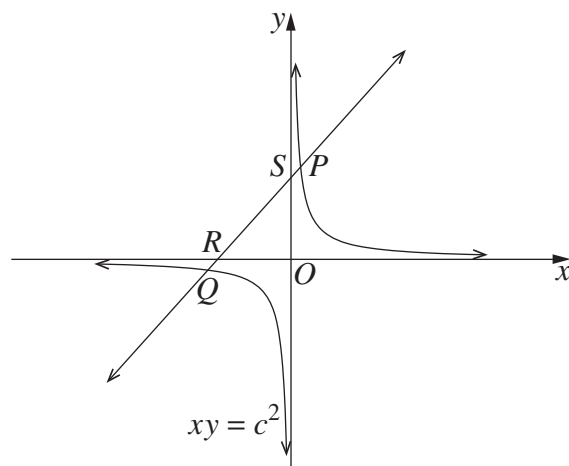
$$\omega^2 \leq \frac{g}{\ell \cos \theta}.$$

Question 13 continues on page 14

Question 13 (continued)

- (d) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$.
The line PQ has equation $x + pqy = c(p + q)$. (Do NOT prove this.)

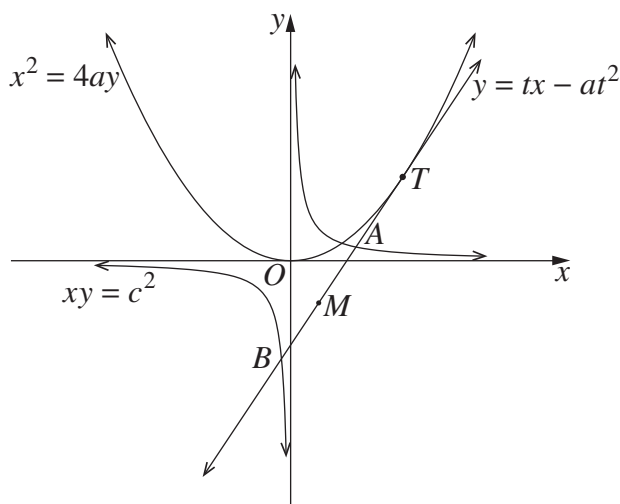
The x and y intercepts of PQ are R and S respectively, as shown in the diagram.



- (i) Show that $PS = QR$.

3

The point $T(2at, at^2)$ lies on the parabola $x^2 = 4ay$. The tangent to the parabola at T intersects the rectangular hyperbola $xy = c^2$ at A and B and has equation $y = tx - at^2$. (Do NOT prove this.) The point M is the midpoint of the interval AB . One such case is shown in the diagram.



- (ii) Using part (i), or otherwise, show that M lies on the parabola $2x^2 = -ay$.

2

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

(a) Using the substitution $t = \tan \frac{\theta}{2}$ evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 - \cos \theta}$. **3**

- (b) A falling particle experiences forces due to gravity and air resistance. The acceleration of the particle is $g - kv^2$, where g and k are positive constants and v is the speed of the particle. (Do NOT prove this.) **3**

Prove that, after falling from rest through a distance, h , the speed of the particle will be $\sqrt{\frac{g}{k}(1 - e^{-2kh})}$.

(c) Let $I_n = \int_{-3}^0 x^n \sqrt{x+3} dx$ for $n = 0, 1, 2, \dots$

- (i) Show that, for $n \geq 1$, **3**

$$I_n = \frac{-6n}{3 + 2n} I_{n-1}.$$

- (ii) Find the value of I_2 . **2**

- (d) Three people, A , B and C , play a series of n games, where $n \geq 2$. In each of the games there is one winner and each of the players is equally likely to win.

- (i) What is the probability that player A wins every game? **1**

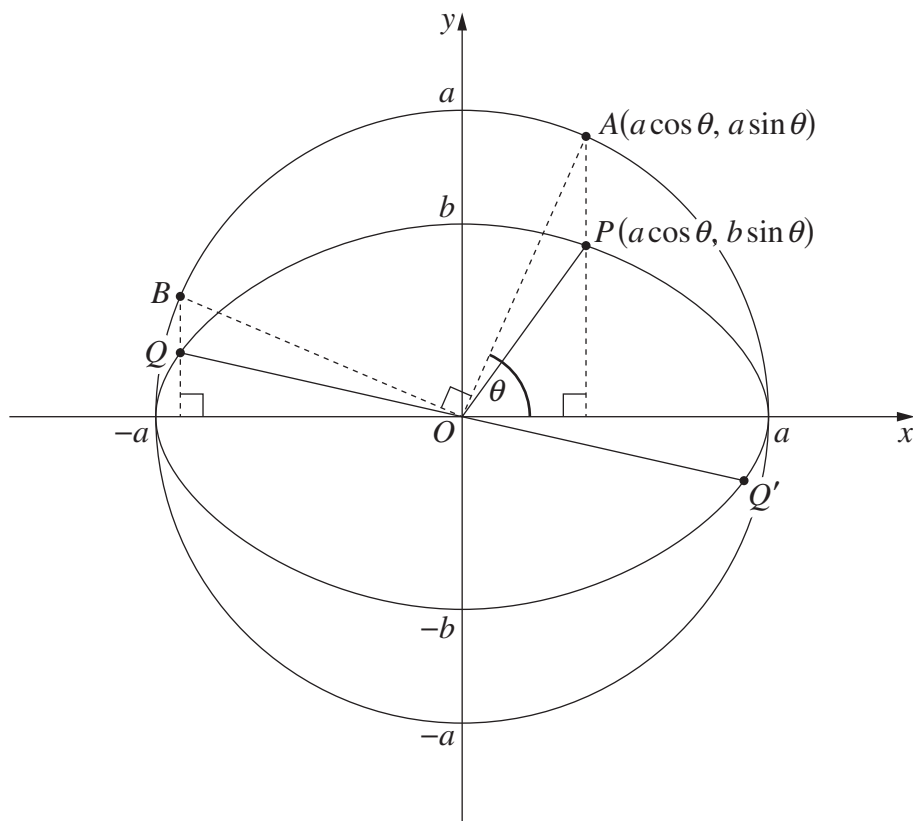
- (ii) Show that the probability that A and B win at least one game each but C never wins, is $\left(\frac{2}{3}\right)^n - 2\left(\frac{1}{3}\right)^n$. **1**

- (iii) Show that the probability that each player wins at least one game is **2**

$$\frac{3^{n-1} - 2^n + 1}{3^{n-1}}.$$

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) The point $P(a \cos \theta, b \sin \theta)$, where $0 < \theta < \frac{\pi}{2}$, lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$. The point $A(a \cos \theta, a \sin \theta)$ lies vertically above P on the auxiliary circle $x^2 + y^2 = a^2$. The point B lies on the auxiliary circle such that $\angle AOB = \frac{\pi}{2}$ and the point Q lies on the ellipse vertically below B , as shown.



- (i) Show that Q has coordinates $(-a \sin \theta, b \cos \theta)$. 2

The line QO meets the ellipse again at $Q'(a \sin \theta, -b \cos \theta)$. (Do NOT prove this.)

- (ii) Show that the minimum size of $\angle POQ'$ is $\tan^{-1}\left(\frac{2ab}{a^2 - b^2}\right)$. 3

Question 15 continues on page 17

Question 15 (continued)

- (b) (i) Use De Moivre's theorem and the expansion of $(\cos \theta + i \sin \theta)^8$ to show that **2**

$$\sin 8\theta = \binom{8}{1} \cos^7 \theta \sin \theta - \binom{8}{3} \cos^5 \theta \sin^3 \theta + \binom{8}{5} \cos^3 \theta \sin^5 \theta - \binom{8}{7} \cos \theta \sin^7 \theta.$$

- (ii) Hence, show that **3**

$$\frac{\sin 8\theta}{\sin 2\theta} = 4(1 - 10 \sin^2 \theta + 24 \sin^4 \theta - 16 \sin^6 \theta).$$

- (c) Let n be a positive integer and let x be a positive real number.

- (i) Show that $x^n - 1 - n(x - 1) = (x - 1)(1 + x + x^2 + \cdots + x^{n-1} - n)$. **1**

- (ii) Hence show that $x^n \geq 1 + n(x - 1)$. **2**

- (iii) Deduce that for positive real numbers a and b , **2**

$$a^n b^{1-n} \geq na + (1 - n)b.$$

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) Use mathematical induction to prove that, for $n \geq 1$,

3

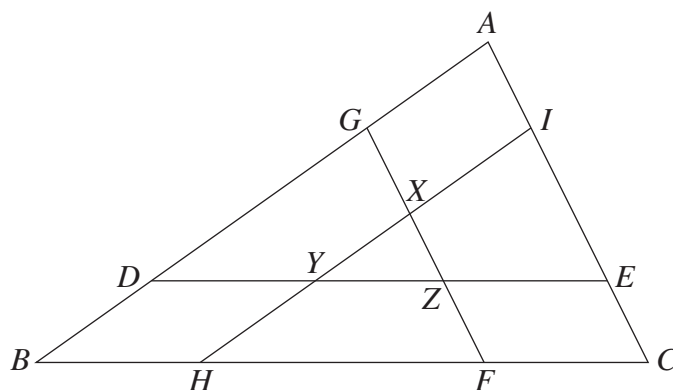
$$x^{(3^n)} - 1 = (x - 1) \left(x^2 + x + 1 \right) \left(x^6 + x^3 + 1 \right) \cdots \left(x^{(2 \times 3^{n-1})} + x^{(3^{n-1})} + 1 \right).$$

- (b) In $\triangle ABC$, point D is chosen on side AB and point E is chosen on side AC so that DE is parallel to BC and $\frac{BC}{DE} = \sqrt{2}$.

The process is repeated two more times. Point F is chosen on BC and point G on BA so that FG is parallel to CA and $\frac{CA}{FG} = \sqrt{2}$.

Point H is chosen on side CB and I on CA so that HI is parallel to BA and $\frac{BA}{HI} = \sqrt{2}$.

The segments FG and HI intersect at X , DE and HI intersect at Y , and DE and FG intersect at Z .



- (i) Prove that $DY = ZE$.

3

- (ii) Find the exact value of the ratio $\frac{YZ}{BC}$.

2

Question 16 continues on page 19

Question 16 (continued)

- (c) Let α, β, γ be the zeros of $p(x) = x^3 + px + q$, where p and q are real and $q \neq 0$.

(i) Show that $(\beta - \gamma)^2 = \alpha^2 + \frac{4q}{\alpha}$. **2**

- (ii) By considering the constant term of a cubic equation with roots $(\alpha - \beta)^2, (\beta - \gamma)^2$ and $(\gamma - \alpha)^2$, or otherwise, show that **3**

$$(\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2 = -(27q^2 + 4p^3).$$

- (iii) Deduce that if $27q^2 + 4p^3 < 0$, then the equation $p(x) = 0$ has 3 distinct real roots. **2**

End of paper

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REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

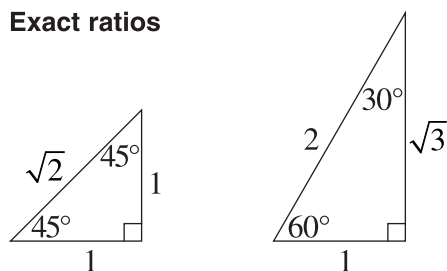
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2018 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	D
4	C
5	A
6	A
7	D
8	B
9	B
10	C

Section II

Question 11 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 zw &= (2 + 3i)(1 - i) \\
 &= 2 - 2i + 3i + 3 \\
 &= 5 + i
 \end{aligned}$$

Question 11 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains $\bar{z}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \bar{z} - \frac{2}{w} &= 2 - 3i - \frac{2}{1 - i} \\
 &= 2 - 3i - \frac{2}{(1 - i)(1 + i)} (1 + i) \\
 &= 2 - 3i - \frac{(2 + 2i)}{1 + 1} \\
 &= 2 - 3i - 1 - i \\
 &= 1 - 4i
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains the value of a , or equivalent merit	2
• Obtains $64 + 16a + b = 0$, or equivalent merit	1

Sample answer:

$$p(x) = x^3 + ax^2 + b$$

$$p(r) = r^3 + ar^2 + b$$

$$p(r) = 0$$

$$r^3 + ar^2 + b = 0 \quad \textcircled{1}$$

$$p'(x) = 3x^2 + 2ax$$

$$p'(4) = 3(4)^2 + 2a(4)$$

$$p'(4) = 0$$

$$48 + 8a = 0$$

$$a = -6$$

$$p(4) = 0$$

$$(4)^3 + a(4)^2 + b = 0$$

$$64 - 6(16) + b = 0$$

$$b = 32$$

Substitute into $\textcircled{1}$

$$r^3 - 6r^2 + 32 = 0$$

$$p(-2) = 0$$

$$r = -2$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	4
• Obtains primitive of the form $p \ln x+1 + q \ln x^2-3 $, or equivalent merit	3
• Obtains the values of a , b and c , or equivalent merit	2
• Obtains the value of a , or equivalent merit	1

Sample answer:

$$\begin{aligned} \frac{x^2 - x - 6}{(x+1)(x^2 - 3)} &= \frac{a(x^2 - 3) + (bx + c)(x + 1)}{(x+1)(x^2 - 3)} \\ &= \frac{ax^2 - 3a + bx^2 + bx + cx + c}{(x+1)(x^2 - 3)} \\ &= \frac{(a+b)x^2 + (b+c)x - 3a + c}{(x+1)(x^2 - 3)} \end{aligned}$$

Equating coefficient

$$a + b = 1, \quad b + c = -1, \quad -3a + c = -6$$

$$b + c = -1 \quad \text{①}$$

$$-3a + c = -6 \quad \text{②}$$

$$\text{①} - \text{②}$$

$$b + 3a = 5 \quad \text{③}$$

$$a + b = 1 \quad \text{④}$$

$$\text{③} - \text{④}$$

$$2a = 4$$

$$a = 2$$

Substitute into ④

$$2 + b = 1$$

$$b = -1$$

Substitute into ①

$$-1 + c = -1$$

$$c = 0$$

$$\int \frac{x^2 - x - 6}{(x+1)(x^2 - 3)} dx$$

$$= \int \frac{2}{x+1} - \frac{x}{x^2 - 3} dx$$

$$= 2 \ln|x+1| - \frac{1}{2} \ln|x^2 - 3| + c$$

Question 11 (d) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$w = i(5 + 2i)$$

$$w = 5i - 2$$

$$w = -2 + 5i$$

Question 11 (d) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$v = u + w$$

$$v = 5 + 2i + -2 + 5i$$

$$v = 3 + 7i$$

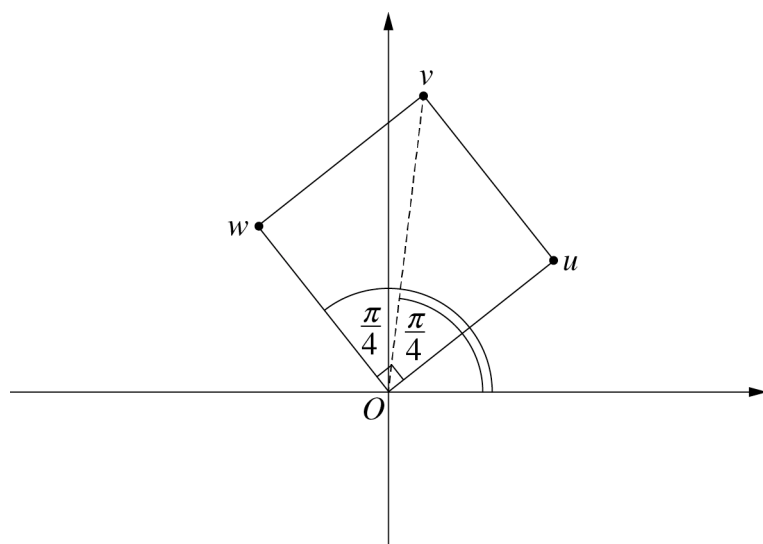
Question 11 (d) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\arg\left(\frac{w}{v}\right) = \arg w - \arg v$$

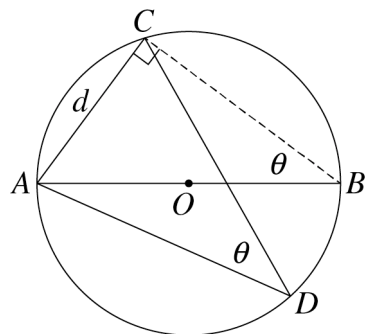
$$= \frac{\pi}{4}$$



Question 11 (e)

Criteria	Marks
• Provides correct explanation	2
• Explains why it is sufficient to consider $\triangle ABC$, or equivalent merit	1

Sample answer:



$$\begin{aligned}\angle ABC &= \angle ADC && \text{(angles standing on same arc are equal)} \\ &= \theta\end{aligned}$$

$$\angle BCA = 90^\circ \quad \text{(angle in a semi-circle is a right angle)}$$

$$\sin \theta = \frac{d}{2r}$$

$$\sin D = \frac{d}{2r}$$

$$d = 2r \sin D$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	3
• Obtains a correct integral for the volume in terms of y , or equivalent merit	2
• Obtains the area of a general slice, or equivalent merit	1

Sample answer:

$$\Delta V = \frac{1}{2} \times x \times x \times \sin 60^\circ \Delta y$$

$$= \frac{\sqrt{3}}{4} x^2 \Delta y$$

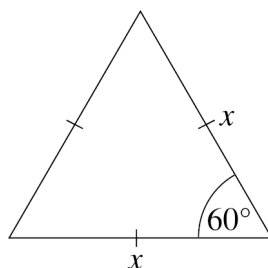
$$V = \int_{-1}^1 \frac{\sqrt{3}}{4} x^2 dy$$

$$= \frac{\sqrt{3}}{2} \int_0^1 (1 - 2y^2 + y^4) dy$$

$$= \frac{\sqrt{3}}{2} \left[y - \frac{2}{3} y^3 + \frac{1}{5} y^5 \right]_0^1$$

$$= \frac{\sqrt{3}}{2} \left(1 - \frac{2}{3} + \frac{1}{5} - 0 \right)$$

$$= \frac{4\sqrt{3}}{15}$$



$$x = 1 - y^2$$

$$x^2 = 1 - 2y^2 + y^4$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly finds derivative of a product using implicit differentiation, or equivalent merit	1

Sample answer:

$$x^2 + xy + y^2 = 3$$

Find $\frac{dy}{dx}$

$$2x + x \frac{dy}{dx} + y + 2y \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} + 2y \frac{dy}{dx} = -2x - y$$

$$\frac{dy}{dx}(x + 2y) = -(2x + y)$$

$$\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains a quadratic from which the solutions can be found, or equivalent merit	1

Sample answer:

$$\frac{dy}{dx} = \frac{-(2x+y)}{x+2y} \quad x^2 + xy + y^2 = 3$$

$$\frac{dy}{dx} = 0 \quad \text{when} \quad -(2x+y) = 0$$

$$2x = -y$$

$$\text{or } y = -2x$$

Substituting into the equation for the curve

$$x^2 + x(-2x) + (-2x)^2 = 3$$

$$x^2 - 2x^2 + 4x^2 = 3$$

$$3x^2 = 3$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\text{when } x = 1 \quad y = -2$$

$$\text{when } x = -1 \quad y = 2$$

Points on the curve where $\frac{dy}{dx} = 0$ are $(1, -2)$ and $(-1, 2)$.

Question 12 (c)

Criteria	Marks
• Provides correct solution	3
• Rewrites integrand as $1 - \frac{5}{(x+1)^2 + 4}$, or equivalent merit	2
• Rewrites integrand as $1 - \frac{5}{x^2 + 2x + 5}$, or equivalent merit	1

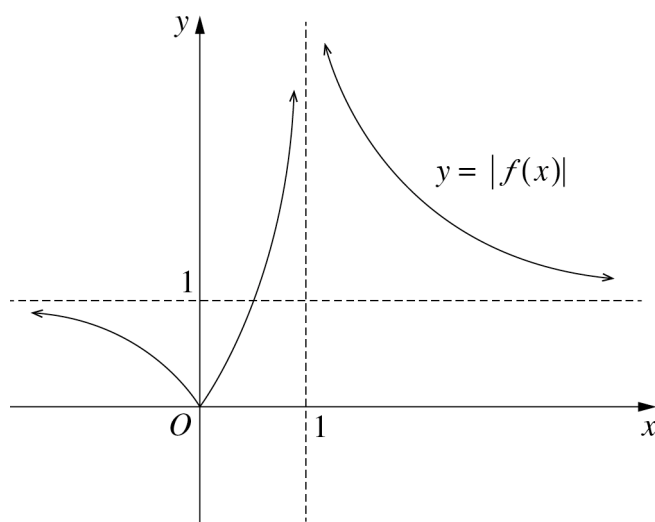
Sample answer:

$$\begin{aligned}
 & \int \frac{x^2 + 2x}{x^2 + 2x + 5} dx \\
 &= \int \frac{x^2 + 2x + 5}{x^2 + 2x + 5} - \frac{5}{x^2 + 2x + 5} dx \\
 &= \int 1 - \frac{5}{(x+1)^2 + 4} dx \\
 &= x - \frac{5}{2} \tan^{-1} \frac{(x+1)}{2} + c
 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct sketch	1

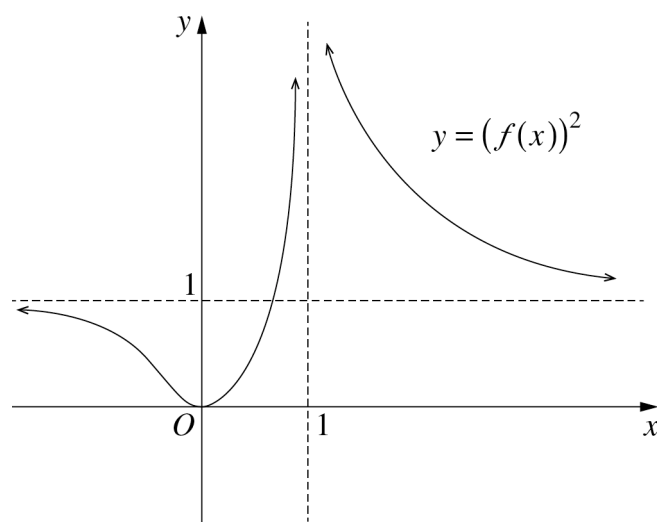
Sample answer:



Question 12 (d) (ii)

Criteria	Marks
• Provides correct sketch	2
• Shows minimum turning point at the origin or equivalent merit	1

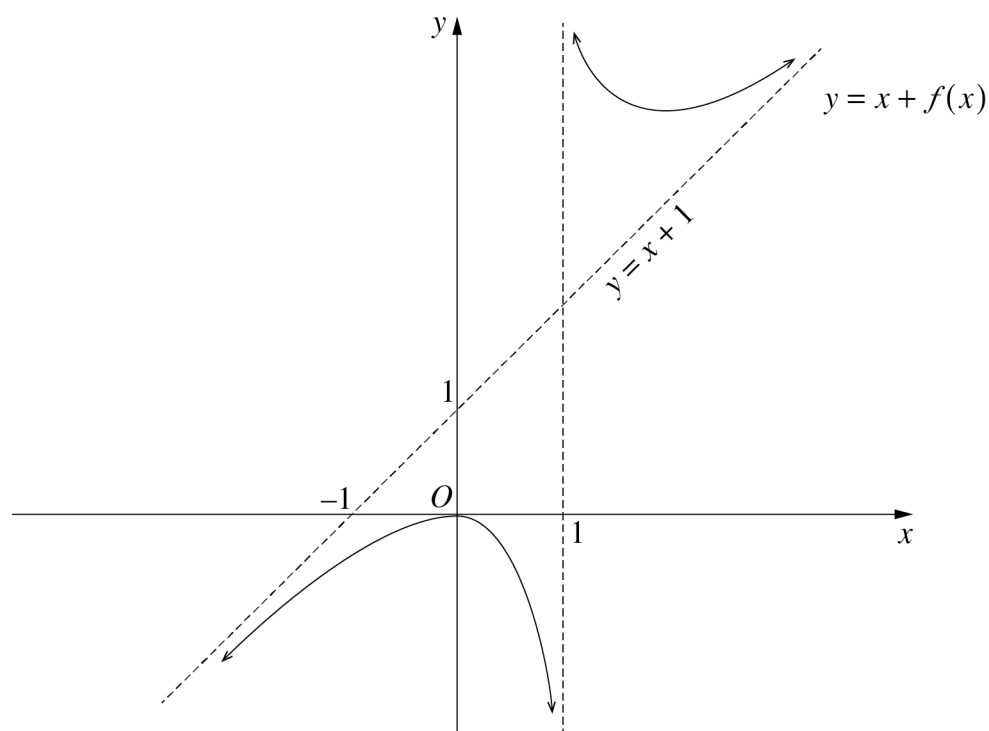
Sample answer:



Question 12 (d) (iii)

Criteria	Marks
• Provides correct sketch	2
• Provides sketch of curve with correct shape, or equivalent merit	1

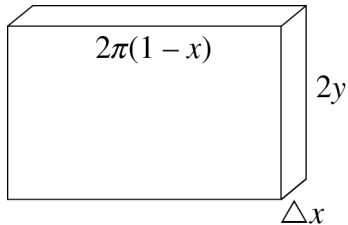
Sample answer:



Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Obtains a correct integral for the volume, or equivalent merit	2
• Correctly finds dV , or equivalent merit	1

Sample answer:



$$V = 4\pi \int_0^1 (1-x) \cdot \sqrt{x}(1-x) dx$$

$$V = 4\pi \int_0^1 \sqrt{x}(1-2x+x^2) dx$$

$$V = 4\pi \int_0^1 x^{\frac{1}{2}} - 2x^{\frac{3}{2}} + x^{\frac{5}{2}} dx$$

$$V = 4\pi \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{4}{5}x^{\frac{5}{2}} + \frac{2}{7}x^{\frac{7}{2}} \right]_0^1$$

$$V = 4\pi \left[\left(\frac{2}{3}(1)^{\frac{3}{2}} - \frac{4}{5}(1)^{\frac{5}{2}} + \frac{2}{7}(1)^{\frac{7}{2}} \right) - 0 \right]$$

$$\text{Volume} = \frac{64\pi}{105} \text{ u}^3$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains $ z ^2$ in terms of 2θ , or equivalent merit	1

Sample answer:

$$\begin{aligned}
 |z| &= \sqrt{(1 - \cos 2\theta)^2 + (\sin 2\theta)^2} \\
 &= \sqrt{1 - 2\cos 2\theta + \cos^2 2\theta + \sin^2 2\theta} \\
 &= \sqrt{2 - 2\cos 2\theta} \\
 &= \sqrt{2(1 - \cos 2\theta)} \\
 &= \sqrt{4\sin^2 \theta} \\
 &= 2\sin \theta \text{ as required}
 \end{aligned}$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds $\tan(\arg(z))$ in terms of $\sin \theta$ and $\cos \theta$, or equivalent merit	1

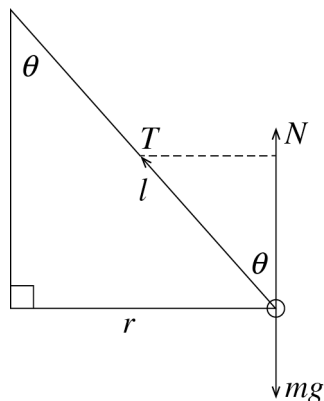
Sample answer:

$$\begin{aligned}
 \arg(z) &= \tan^{-1} \left(\frac{\sin 2\theta}{1 - \cos 2\theta} \right) \\
 &= \tan^{-1} \left(\frac{2\sin \theta \cos \theta}{2\sin^2 \theta} \right) \\
 &= \tan^{-1} \left(\frac{\cos \theta}{\sin \theta} \right) \\
 &= \tan^{-1}(\cot \theta) \\
 &= \frac{\pi}{2} - \theta
 \end{aligned}$$

Question 13 (c)

Criteria	Marks
• Provides correct solution	3
• Obtains an expression for N independent of T , or equivalent merit	2
• Obtains an equation for the vertical components of the forces, or equivalent merit	1

Sample answer:



$$T \cos \theta + N = mg \quad \text{①}$$

$$T \sin \theta = mr\omega^2 \quad \text{②}$$

$$T = \frac{mr\omega^2}{\sin \theta}$$

Substitute into ①

$$\frac{mr\omega^2}{\sin \theta} \cdot \cos \theta + N = mg$$

$$\frac{mr\omega^2 \cdot l}{r} \cdot \cos \theta + N = mg$$

$$N = mg - m\omega^2 l \cos \theta$$

$N \geq 0$ to remain in contact

$$mg - m\omega^2 l \cos \theta \geq 0$$

$$m\omega^2 l \cos \theta \leq mg$$

$$\omega^2 \leq \frac{mg}{ml \cos \theta}$$

$$\omega^2 \leq \frac{g}{l \cos \theta}$$

Question 13 (d) (i)

Criteria	Marks
• Provides correct solution	3
• Correctly finds the distance PS or equivalent merit	2
• Correctly finds the coordinates of R and S , or equivalent merit	1

Sample answer:

$$P\left(cp, \frac{c}{p}\right), \quad S\left(0, \frac{c(p+q)}{pq}\right)$$

$$PS = \sqrt{(0 - cp)^2 + \left(\frac{c(p+q)}{pq} - \frac{c}{p}\right)^2}$$

$$PS = \sqrt{c^2 p^2 + \left(\frac{cp + cq - cq}{pq}\right)^2}$$

$$PS = \sqrt{c^2 p^2 + \frac{c^2 p^2}{p^2 q^2}}$$

$$PS = \sqrt{c^2 \left(p^2 + \frac{1}{q^2}\right)}$$

$$PS = c \sqrt{p^2 + \frac{1}{q^2}}$$

$$Q\left(cq, \frac{c}{q}\right), \quad R(c(p+q), 0)$$

$$QR = \sqrt{(cp + cq - cq)^2 + \left(0 - \frac{c}{q}\right)^2}$$

$$QR = \sqrt{c^2 p^2 + \frac{c^2}{q^2}}$$

$$QR = \sqrt{c^2 \left(p^2 + \frac{1}{q^2}\right)}$$

$$= c \sqrt{p^2 + \frac{1}{q^2}}$$

$$= PS$$

Question 13 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds M , or equivalent merit	1

Sample answer:

$$y = tx - at^2$$

$$\text{At } x = 0, y = -at^2 \quad \therefore \text{y-intercept is } (0, -at^2)$$

$$\begin{aligned} \text{At } y = 0, 0 &= tx - at^2 \\ x &= at \quad \therefore \text{x-intercept is } (at, 0) \end{aligned}$$

Using part (i), the midpoint of AB will also be the midpoint of the x and y -intercepts.

$$\begin{aligned} \therefore M \text{ is } &\left(\frac{0 + at}{2}, \frac{-at^2 + 0}{2} \right) \\ &\text{ie } \left(\frac{at}{2}, \frac{-at^2}{2} \right) \end{aligned}$$

substitute M into $2x^2 = -ay$

$$\begin{aligned} \text{LHS} &= 2 \times \left(\frac{at}{2} \right)^2 \\ &= \frac{2a^2t^2}{4} \\ &= \frac{a^2t^2}{2} \\ &= -a \left(-\frac{at^2}{2} \right) \\ &= -ay \\ &= \text{RHS} \end{aligned}$$

$\therefore M$ lies on the parabola $2x^2 = -ay$

Question 13 (d) (ii)

Alternative solution

$$y = tx - at^2 \quad \textcircled{1}$$

$$xy = c^2 \quad \textcircled{2}$$

$$y = \frac{c^2}{x}$$

Substitute into ①

$$\frac{c^2}{x} = tx - at^2$$

$$c^2 = tx^2 - at^2x$$

$$tx^2 - at^2x - c^2 = 0$$

$$x = \frac{at^2 \pm \sqrt{(-at^2)^2 - 4t(-c^2)}}{2t}$$

$$x = \frac{at^2 \pm \sqrt{a^2t^4 + 4c^2t}}{2t}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$M = \left(\frac{2at^2}{4t}, ? \right)$$

$$M = \left(\frac{at}{2}, ? \right)$$

$$x = \frac{at}{2} \text{ lies on } y = tx - at^2$$

$$y = t \cdot \frac{at}{2} - at^2$$

$$y = -\frac{at^2}{2}$$

$$\therefore M = \left(\frac{at}{2}, -\frac{at^2}{2} \right)$$

$$x = \frac{at}{2}, \quad y = -\frac{at^2}{2}$$

$$t = \frac{2x}{a}, \quad y = -\frac{a}{2} \left(\frac{2x}{a} \right)^2$$

$$y = -\frac{a}{2} \left(\frac{4x^2}{a^2} \right)$$

$$y = -\frac{2x^2}{a}$$

$$2x^2 = -ay$$

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral, or equivalent merit	2
• Correctly obtains $d\theta$, or equivalent merit	1

Sample answer:

$$I = \int_0^{\frac{\pi}{2}} \frac{d\theta}{2 - \cos\theta}$$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 - \frac{1-t^2}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \int_0^1 \frac{2 \cdot dt}{2(1+t^2) - (1-t^2)}$$

$$= \int_0^1 \frac{2dt}{1+3t^2}$$

$$= \frac{2}{3} \int_0^1 \frac{dt}{\frac{1}{3} + t^2}$$

$$= \frac{2}{3} \cdot \frac{1}{\frac{1}{\sqrt{3}}} \left[\tan^{-1} \frac{t}{\frac{1}{\sqrt{3}}} \right]_0^1$$

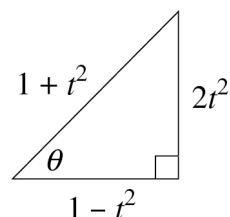
$$= \frac{2\sqrt{3}}{3} \left[\tan^{-1} \sqrt{3}t \right]_0^1$$

$$= \frac{2\sqrt{3}}{3} (\tan^{-1} \sqrt{3} - \tan^{-1} 0)$$

$$= \frac{2\sqrt{3}}{3} \cdot \frac{\pi}{3}$$

$$= \frac{2\sqrt{3}}{9} \pi$$

$$\text{Let } t = \tan \frac{\theta}{2}$$



$$dt = \frac{1}{2} \sec^2 \frac{\theta}{2} \cdot d\theta$$

$$2dt = (1+t^2)d\theta$$

$$d\theta = \frac{2dt}{1+t^2}$$

$$\theta = 0, \quad t = 0$$

$$\theta = \frac{\pi}{2}, \quad t = 1$$

Question 14 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains x , or equivalent merit	2
• Obtains $\frac{dx}{dv} = \frac{v}{g - kv^2}$, or equivalent merit	1

Sample answer:

$$\ddot{x} = g - kv^2$$

$$v \cdot \frac{dv}{dx} = g - kv^2$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$x = \int \frac{v}{g - kv^2} dv$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + c$$

$$x = 0, v = 0 \Rightarrow 0 = -\frac{1}{2k} \ln(g - k(0)^2) + C$$

$$\Rightarrow c = \frac{1}{2k} \ln(g)$$

$$\therefore x = -\frac{1}{2k} \ln(g - kv^2) + \frac{1}{2k} \ln(g)$$

$$= -\frac{1}{2k} \ln\left(\frac{g - kv^2}{g}\right)$$

$$= -\frac{1}{2k} \ln\left(1 - \frac{k}{g} v^2\right)$$

$$\text{Let } x = h \Rightarrow -2kh = \ln\left(1 - \frac{k}{g} v^2\right)$$

$$1 - \frac{k}{g} v^2 = e^{-2kh}$$

$$1 - e^{-2kh} = \frac{k}{g} v^2$$

$$v^2 = \frac{g}{k} (1 - e^{-2kh})$$

$$v = \sqrt{\frac{g}{k} (1 - e^{-2kh})}$$

$$v > 0$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Correctly applies integration by parts, or equivalent merit	2
• Attempts to apply integration by parts or equivalent merit	1

Sample answer:

$$I_n = \int_{-3}^0 x^n \sqrt{x+3} \, dx \quad n = 0, 1, 2, \dots$$

$$\text{Let } u = x^n \quad v^1 = (x+3)^{\frac{1}{2}}$$

$$u^1 = nx^{n-1} \quad v = \frac{2(x+3)^{\frac{3}{2}}}{3}$$

$$I_n = \left[\frac{2x^n(x+3)^{\frac{3}{2}}}{3} \right]_{-3}^0 - \int_{-3}^0 \frac{2nx^{n-1}(x+3)^{\frac{3}{2}}}{3} \, dx$$

$$I_n = (0-0) - \frac{-2n}{3} \int_{-3}^0 x^{n-1}(x+3)\sqrt{x+3} \, dx$$

$$I_n = -\frac{2n}{3} \int_{-3}^0 (x^n \sqrt{x+3} + 3x^{n-1} \sqrt{x+3}) \, dx$$

$$I_n = -\frac{2n}{3} \int_{-3}^0 x^n \sqrt{x+3} \, dx - 2n \int_{-3}^0 x^{n-1} \sqrt{x+3} \, dx$$

$$I_n = -\frac{2n}{3} I_n - 2n I_{n-1}$$

$$I_n \left(1 + \frac{2n}{3} \right) = -2n I_{n-1}$$

$$I_n \left(\frac{3+2n}{3} \right) = -2n I_{n-1}$$

$$I_n = \frac{-6n}{3+2n} I_{n-1}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds I_2 in terms of I_0 , or equivalent merit	1

Sample answer:

$$I_2 = \frac{-12}{7} I_1$$

$$= \frac{-12}{7} \times \frac{-6}{5} I_0$$

$$I_0 = \int_{-3}^0 \sqrt{x+3} dx$$

$$= \left[\frac{2(x+3)^{\frac{3}{2}}}{3} \right]_{-3}^0$$

$$= \frac{2}{3} (3)^{\frac{3}{2}} - 0$$

$$= \frac{2 \times 3\sqrt{3}}{3}$$

$$= 2\sqrt{3}$$

$$\therefore I_2 = \frac{72}{35} \times 2\sqrt{3}$$

$$= \frac{144\sqrt{3}}{35}$$

Question 14 (d) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$P(A \text{ wins}) = \left(\frac{1}{3}\right)^n$$

Question 14 (d) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$P(C \text{ never wins}) = \left(\frac{2}{3}\right)^n$$

This includes when A wins all and B wins all.

$$P(A \text{ wins all games or } B \text{ wins all games}) = 2 \times \left(\frac{1}{3}\right)^n$$

$\therefore$ Required probability where A and B both win a game is: $\left(\frac{2}{3}\right)^n - 2\left(\frac{1}{3}\right)^n$

Question 14 (d) (iii)

Criteria	Marks
• Provides correct solution	2
• Correctly deals with the case where one player wins all the games, or equivalent merit	1

Sample answer:

If A , B or C wins all the games

$$\text{Probability is: } 3 \times \left(\frac{1}{3}\right)^n = \frac{1}{3^{n-1}}$$

If two people win at least one game but not the third person

$$\begin{aligned} \text{Probability is: } & 3 \left[\left(\frac{2}{3}\right)^n - 2\left(\frac{1}{3}\right)^n \right] \\ &= \frac{2^n - 2}{3^{n-1}} \end{aligned}$$

$P(\text{each player wins at least one game})$

$$\begin{aligned} &= 1 - \left[\frac{1}{3^{n-1}} + \frac{2^n - 2}{3^{n-1}} \right] \\ &= \frac{3^{n-1} - 1 - 2^n + 2}{3^{n-1}} \\ &= \frac{3^{n-1} - 2^n + 1}{3^{n-1}} \end{aligned}$$

Question 15 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Finds coordinates of B , or equivalent merit	1

Sample answer:

Since $\angle AOB = \frac{\pi}{2}$

B has coordinates $\left(a \cos\left(\theta + \frac{\pi}{2}\right), a \sin\left(\theta + \frac{\pi}{2}\right) \right)$
ie $(-a \sin \theta, a \cos \theta)$

Q has same x-coordinate

$$\begin{aligned} \frac{y^2}{b^2} &= 1 - \frac{x^2}{a^2} \\ &= 1 - \frac{a^2 \sin^2 \theta}{a^2} \\ &= \cos^2 \theta \end{aligned}$$

$$\therefore y^2 = b^2 \cos^2 \theta$$

$$y = b \cos \theta \quad (\text{since } y > 0)$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds the minimum value of $\tan(\angle POQ')$, or equivalent merit	2
• Finds an expression for $\tan(\angle POQ')$, or equivalent merit	1

Sample answer:

$$M_{OP} = \frac{b \sin \theta}{a \cos \theta} \quad (\cos \theta \neq 0)$$

$$M_{OQ'} = -\frac{b \cos \theta}{a \sin \theta} \quad (\sin \theta \neq 0)$$

$$\begin{aligned} \therefore \tan \angle POQ' &= \left| \frac{\frac{b \sin \theta}{a \cos \theta} - \left(-\frac{b \cos \theta}{a \sin \theta}\right)}{1 + \frac{b \sin \theta}{a \cos \theta} \left(-\frac{b \cos \theta}{a \sin \theta}\right)} \right| \\ &= \left| \frac{ab \sin^2 \theta + ab \cos^2 \theta}{a^2 \sin \theta \cos \theta - b^2 \sin \theta \cos \theta} \right| \end{aligned}$$

(Multiplying numerator and denominator by $a^2 \sin \theta \cos \theta$)

$$= \frac{2ab}{(a^2 - b^2) \sin 2\theta} \quad (\text{all terms are positive so absolute value is not necessary})$$

This expression is least when $\sin 2\theta$ is greatest
ie when $\sin 2\theta = 1$

Result follows

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly uses binomial theorem, or equivalent merit	1

Sample answer:

$$(\cos \theta + i \sin \theta)^8 = \cos 8\theta + i \sin 8\theta \quad (\text{De Moivre})$$

$$\begin{aligned} \text{Also } (\cos \theta + i \sin \theta)^8 &= \cos^8 \theta + \binom{8}{1} \cos^7 \theta i \sin \theta + \binom{8}{2} \cos^6 \theta i^2 \sin^2 \theta + \binom{8}{3} \cos^5 \theta i^3 \sin^3 \theta + \\ &\quad \binom{8}{4} \cos^4 \theta i^4 \sin^4 \theta + \dots + i^8 \sin^8 \theta \\ &= \cos^8 \theta \binom{8}{1} \cos^7 \theta \sin \theta i - \binom{8}{2} \cos^6 \theta \sin^2 \theta - \binom{8}{3} \cos^5 \theta \sin^3 \theta i + \\ &\quad \binom{8}{4} \cos^4 \theta \sin^4 \theta + \dots + \sin^8 \theta \end{aligned}$$

Equating imaginary parts

$$\sin 8\theta = \binom{8}{1} \cos^7 \theta \sin \theta - \binom{8}{3} \cos^5 \theta \sin^3 \theta + \binom{8}{5} \cos^3 \theta \sin^5 \theta - \binom{8}{7} \cos \theta \sin^7 \theta$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Obtains an expression for $\frac{\sin 8\theta}{\sin 2\theta}$ in terms of $\sin \theta$, or equivalent merit	2
• Correctly divides by $\sin 2\theta$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{\sin 8\theta}{\sin 2\theta} &= \frac{\binom{8}{1}\cos^7\theta\sin\theta - \binom{8}{3}\cos^5\theta\sin^3\theta + \binom{8}{5}\cos^3\theta\sin^5\theta - \binom{8}{7}\cos\theta\sin^7\theta}{2\sin\theta\cos\theta} \\
 &= \frac{1}{2} \left[\binom{8}{1}\cos^6\theta - \binom{8}{3}\cos^4\theta\sin^2\theta + \binom{8}{5}\cos^2\theta\sin^4\theta - \binom{8}{7}\sin^6\theta \right] \\
 &= \frac{1}{2} \left[\binom{8}{1}(1-\sin^2\theta)^3 - \binom{8}{3}(1-\sin^2\theta)^2\sin^2\theta + \binom{8}{5}(1-\sin^2\theta)^2\sin^4\theta - \binom{8}{7}\sin^6\theta \right] \\
 &= \frac{1}{2} \left[8(1-3\sin^2\theta+3\sin^4\theta-\sin^6\theta) - 56(\sin^2\theta-2\sin^4\theta+\sin^6\theta) \right. \\
 &\quad \left. + 56(\sin^4\theta-\sin^6\theta) - 8\sin^6\theta \right] \\
 &= 4 \left[1-3\sin^2\theta+3\sin^4\theta-\sin^6\theta - 7\sin^2\theta+14\sin^4\theta-7\sin^6\theta \right. \\
 &\quad \left. + 7\sin^4\theta-7\sin^6\theta-\sin^6\theta \right] \\
 &= 4 \left[1-10\sin^2\theta+24\sin^4\theta-16\sin^6\theta \right]
 \end{aligned}$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Show that $x^n - 1 - n(x - 1) = (x - 1)(1 + x + \dots + x^{n-1} - n)$

$x > 0, \quad x \in IR$

$$\begin{aligned}
 (x^n - 1) - n(x - 1) &= (x - 1)(x^{n-1} + x^{n-2} + \dots + x^2 + x + 1) - n(x - 1) \\
 &= (x - 1)(x^{n-1} + x^{n-2} + \dots + x^2 + x + 1 - n) \\
 &= (x - 1)(1 + x + x^2 + \dots + x^{n-1} - n)
 \end{aligned}$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Recognises the two cases $x \geq 1$, $x < 1$, or equivalent merit	1

Sample answer:

Hence show that $x^n \geq 1 + n(x - 1)$

When $x = 1$, $x^n = 1$, and $1 + n(x - 1) = 1$

so $x^n \geq 1 + n(x - 1)$

Suppose $0 < x < 1$ so $0 < x^k < 1$ for all integers $k > 0$

$(x - 1) < 0$ and $1 + x + x^2 + \dots + x^{n-1} < 1 + 1 + \dots + 1$ (n terms)

so $1 + x + x^2 + \dots + x^{n-1} < n$

so $1 + x + x^2 + \dots + x^{n-1} - n < 0$

when $x < 1$ $(x - 1)(1 + x + x^2 + \dots + x^{n-1} - n) > 0$

suppose $x > 1$ $x^k > 1$ for all integers $k > 0$

$(x - 1) > 0$ $1 + x + x^2 + \dots + x^{n-1} > n$

so $1 + x + x^2 + \dots + x^{n-1} - n > 0$

so when $x > 1$ $(x - 1)(1 + x + x^2 + \dots + x^{n-1} - n) > 0$

Thus

$$x^n - 1 - n(x - 1) \geq 0$$

so $x^n \geq 1 + n(x - 1)$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Considers $x = \frac{a}{b}$, or equivalent merit	1

Sample answer:

Deduce that for positive real numbers

a and b

$$a^n b^{1-n} \geq na + (1-n)b$$

Let $x = \frac{a}{b} > 0$ and use part (ii)

$$\frac{a^n}{b^n} \geq 1 + n\left(\frac{a}{b} - 1\right)$$

$$a^n b^{-n} \geq \frac{b}{b} + \frac{na}{b} - \frac{bn}{b}$$

$$a^n b^{-n} \times b \geq na + b - bn$$

$$a^n b^{1-n} \geq na + b(1-n)$$

Question 16 (a)

Criteria	Marks
• Provides correct proof	3
• Establishes the inductive step, or equivalent merit	2
• Shows true for $n = 1$, or equivalent merit	1

Sample answer:

Prove for $n \geq 1$

$$x^{3^n} - 1 = (x - 1)(x^2 + x + 1)(x^6 + x^3 + 1) \dots (x^{2 \times 3^{n-1}} + x^{3^{n-1}} + 1)$$

When $n = 1$

$$\text{LHS} = x^3 - 1 = (x - 1)(x^2 + x + 1) \text{ as required} = \text{RHS}$$

Assume that the proposition is true when $n = k$

$$\text{so } x^{3^k} - 1 = (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^{k-1}} + x^{3^{k-1}} + 1)$$

Show the proposition is true when $n = k + 1$

$$\begin{aligned} x^{3^{k+1}} - 1 &= x^{3^k \cdot 3} - 1 \\ &= (x^{3^k})^3 - 1 \\ &= (x^{3^k} - 1) \left((x^{3^k})^2 + (x^{3^k}) + 1 \right) \\ &= (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^{k-1}} + x^{3^{k-1}} + 1) (x^{2 \times 3^k} + x^{3^k} + 1) \end{aligned}$$

Since proposition is true for $n = k + 1$

$$\text{By POMI } x^{3^n} - 1 = (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^{n-1}} + x^{3^{n-1}} + 1)$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct proof	3
• Proves that $BF = HC$, or equivalent merit	2
• Observes that $BH = DY$, or equivalent merit	1

Sample answer:

In triangles BGF and BAC $\angle B$ is common

As $GF \parallel AC$

$\angle GFB = \angle ACB$ (corres)

so $\triangle BGF$ is similar to $\triangle BAC$

$$\frac{AC}{GF} = \sqrt{2}$$

so $\frac{BC}{BF} = \sqrt{2}$

similarly $\frac{BC}{HC} = \sqrt{2}$

hence $BF = HC$

$AB \parallel IH$ and $BC \parallel DE$

so $DYHB$ is a parallelogram

similarly $ZECF$ is a parallelogram

hence

$$\begin{aligned}
 DY &= BH && \text{(opposite sides in parallelogram)} \\
 &= BC - HC \\
 &= BC - BF \\
 &= FC \\
 &= ZE && \text{(opposite sides in parallelogram)}
 \end{aligned}$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains an expression for YZ in terms of BC , or equivalent merit	1

Sample answer:

$$YZ = DE - DY - ZE$$

$$= DE - 2DY \quad (\text{from (i)})$$

$$= \frac{BC}{\sqrt{2}} - 2(BC - HC)$$

$$= \frac{BC}{\sqrt{2}} - 2BC + 2HC$$

$$= \frac{BC}{\sqrt{2}} - 2BC + 2\frac{BC}{\sqrt{2}}$$

$$= 3\frac{BC}{\sqrt{2}} - 2BC$$

$$= BC \left(\frac{3 - 2\sqrt{2}}{\sqrt{2}} \right)$$

$$\therefore \frac{YZ}{BC} = \frac{3 - 2\sqrt{2}}{\sqrt{2}}$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $\alpha + \beta + \gamma = 0$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 (\beta - \gamma)^2 &= \beta^2 - 2\beta\gamma + \gamma^2 \\
 &= \beta^2 + 2\beta\gamma + \gamma^2 - 4\beta\gamma \\
 &= (\beta + \gamma)^2 - 4\beta\gamma \\
 &= (\alpha + \beta + \gamma - \alpha)^2 - \frac{4\alpha\beta\gamma}{\alpha} \\
 &= (0 - \alpha)^2 - \frac{4(-q)}{\alpha}
 \end{aligned}$$

(since $\alpha + \beta + \gamma = 0$ and $\alpha\beta\gamma = -q$)

$$= \alpha^2 + \frac{4q}{\alpha}$$

Question 16 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Obtains cubic equation in y having noted $y + p \neq 0$, or equivalent merit	2
• Lets $y = x^2 + \frac{4q}{x}$, or equivalent merit	1

Sample answer:

We want to form an equation with roots

$$\alpha^2 + \frac{4q}{\alpha}, \quad \beta^2 + \frac{4q}{\beta} \quad \text{and} \quad \gamma^2 + \frac{4q}{\gamma}$$

Suppose x is a solution of the original equation

$$\text{Let } y = x^2 + \frac{4q}{x}$$

$$\text{then } xy = x^3 + 4q$$

$$= -px - q + 4q \quad \text{since } x^3 + px + q = 0$$

$$x(y + p) = 3q$$

Since $q \neq 0$, it follows $y + p \neq 0$

$$\therefore x = \frac{3q}{y + p}$$

Since x satisfies original equation

$$\left(\frac{3q}{y + p} \right)^3 + p \left(\frac{3q}{y + p} \right) + q = 0$$

After multiplying by $(y + p)^3$

$$27q^2 + 3pq(y + p)^2 + q(y + p)^3 = 0$$

This is an equation with roots $(\alpha - \beta)^2$, $(\beta - \gamma)^2$ and $(\gamma - \alpha)^2$

$$\text{Product of these roots} = \frac{-\text{constant term}}{\text{leading coefficient}}$$

$$= -\frac{27q^3 + 3pq \cdot p^2 + qp^3}{q} \quad (q \neq 0)$$

$$= -(27q^2 + 3p^3 + p^3)$$

$$= -(27q^2 + 4p^3)$$

Question 16 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Observes that $p(x)$ has distinct roots which are all real or 1 real and 2 complex, or equivalent merit	1

Sample answer:

Suppose $27q^2 + 4p^3 < 0$

Then $(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2 > 0$

The roots α, β and γ are distinct otherwise at least one factor would be zero.

If not all the roots are real then there must be one real root and two roots which are complex conjugates (since coefficients of the cubic are real).

Let the roots be $\alpha, a + ib, a - ib$ where α, a, b are real

$$(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2$$

let α be the real root and $\beta = a + ib$ and $\gamma = a - ib$ where a, b are real

$$(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2 = (\alpha - \beta)^2(\beta - \gamma)^2(\alpha - \gamma)^2$$

$$= \left[(\alpha - a) - ib \right]^2 \left[2ib \right]^2 \left[(\alpha - a) + ib \right]^2$$

$$= \left(\left[(\alpha - a) - ib \right] \left[(\alpha - a) + ib \right] \right)^2 \times -4b^2$$

$$= -4b^2 \left[(\alpha - a)^2 + b^2 \right]^2$$

LHS > 0 RHS < 0 contradictions

Hence all roots are real.

2018 HSC Mathematics Extension 2 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	4.1	E8
2	1	3.2	E4
3	1	7.5	E4
4	1	1.5, 1.7	E6
5	1	5.1	E7
6	1	2.4	E3
7	1	2.5	E3
8	1	1.8	E6
9	1	2.1, 8.3	E3
10	1	1.4	E6

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	2.1	E3
11 (a) (ii)	2	2.1	E3
11 (b)	3	7.2	E4
11 (c)	4	4.1, 7.6	E8
11 (d) (i)	1	2.2	E3
11 (d) (ii)	1	2.2	E3
11 (d) (iii)	1	2.2	E3
11 (e)	2	8.1	E2
12 (a)	3	5.1	E7
12 (b) (i)	2	1.8	E6
12 (b) (ii)	2	1.8	E6
12 (c)	3	4.1	E8
12 (d) (i)	1	1.3	E6
12 (d) (ii)	2	1.6	E6
12 (d) (iii)	2	1.2	E6
13 (a)	3	5.1	E7
13 (b) (i)	2	2.2	E3
13 (b) (ii)	2	2.2	E3
13 (c)	3	6.3.1, 6.3.2	E5
13 (d) (i)	3	3.3	E3, E4
13 (d) (ii)	2	3.3	E3, E4
14 (a)	3	4.1	E8
14 (b)	3	6.2.3	E5

Question	Marks	Content	Syllabus outcomes
14 (c) (i)	3	4.1	E8
14 (c) (ii)	2	4.1	E8
14 (d) (i)	1	8	E2
14 (d) (ii)	1	8	E2
14 (d) (iii)	2	8	E2
15 (a) (i)	2	3.1	E4
15 (a) (ii)	3	3.1	E4
15 (b) (i)	2	2.4, 8	E3
15 (b) (ii)	3	8	E4
15 (c) (i)	1	8	E2, E9
15 (c) (ii)	2	8.3	E2, E9
15 (c) (iii)	2	8.3	E2, E9
16 (a)	3	8.2	E9
16 (b) (i)	3	8	E9
16 (b) (ii)	2	8	E9
16 (c) (i)	2	7.5	E4
16 (c) (ii)	3	7.5	E4
16 (c) (iii)	2	2.1, 7.5	E4